

120 Days of Quantitative Finance

Week 7: Time Series Analysis Basics

Day 48: Mean Reversion and Random Walk

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1. Random Walk Model

A random walk is defined as:

$$S_t = S_{t-1} + \epsilon_t$$

where:

$$\epsilon_t \sim N(0, \sigma^2)$$

Properties

- Prices are non-stationary.
- Returns are stationary.
- Variance grows over time.

$$\text{Var}(S_t) = t\sigma^2$$

Random walk behavior is consistent with the Efficient Market Hypothesis.

2. Mean Reversion (AR(1) Process)

$$X_t = \phi X_{t-1} + \epsilon_t$$

If:

$$|\phi| < 1$$

the process is stationary and mean-reverting.

Variance

$$Var(X_t) = \frac{\sigma^2}{1 - \phi^2}$$

Shocks decay geometrically:

$$Impact \propto \phi^k$$

3. Continuous-Time Mean Reversion

The Ornstein-Uhlenbeck process:

$$dX_t = \kappa(\mu - X_t)dt + \sigma dW_t$$

where:

- κ = speed of reversion
- μ = long-run mean
- σ = volatility

The drift pulls the process toward μ .

4. Statistical Identification

Random Walk:

- Unit root present
- ADF test fails to reject null

Mean Reversion:

- No unit root
- ADF rejects null

5. Financial Implications

Random Walk:

- Limited predictability
- Consistent with weak-form efficiency

Mean Reversion:

- Statistical arbitrage opportunities
- Z-score based strategies
- Spread trading

Conclusion

Understanding whether a process follows a random walk or mean reversion is fundamental for building quantitative trading strategies, risk models, and forecasting frameworks.